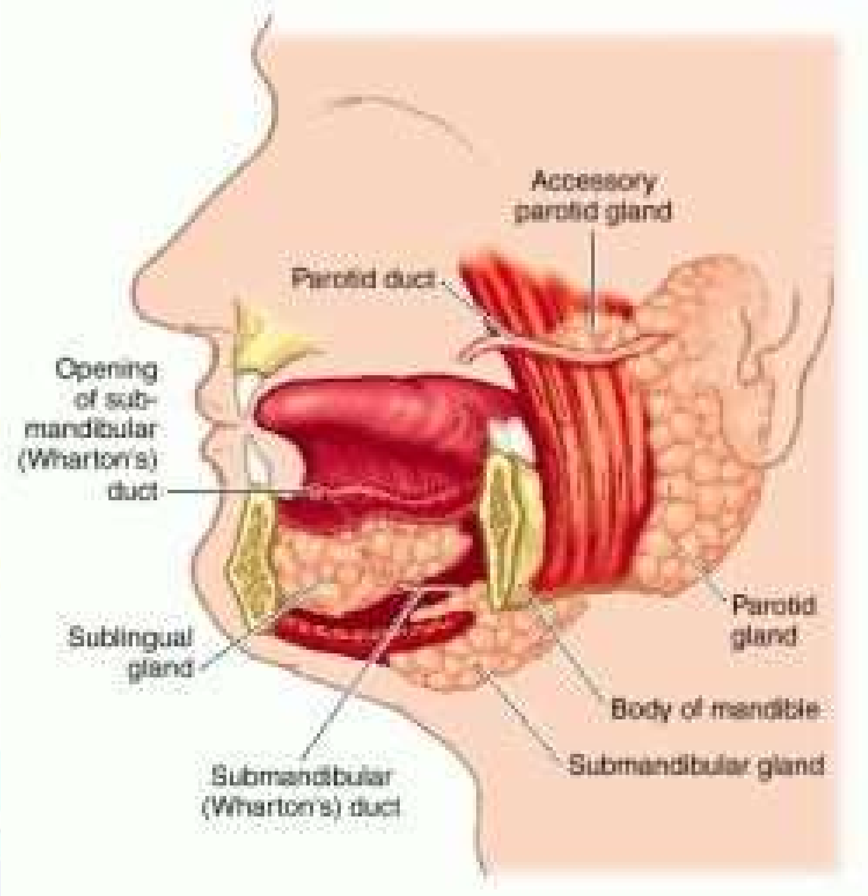
Salivary glands

Learning objectives

- At the end of the lecture the student should be able to know:
- Salivary glands and their location
- Histology of parotid gland
- Histology of submandibular gland
- Histology of sublingual gland.

Salivary glands

- **Salivary glands** are exocrine glands, glands with ducts, that produce saliva.
- **The Major Salivary Glands**
  - **Parotid**
  - **Submandibular**
  - **Sublingual**
  - **The Minor Salivary Glands**
- **They differ from one another in the relative abundance of serous and mucous acini, and in the length of the various kinds of ducts**

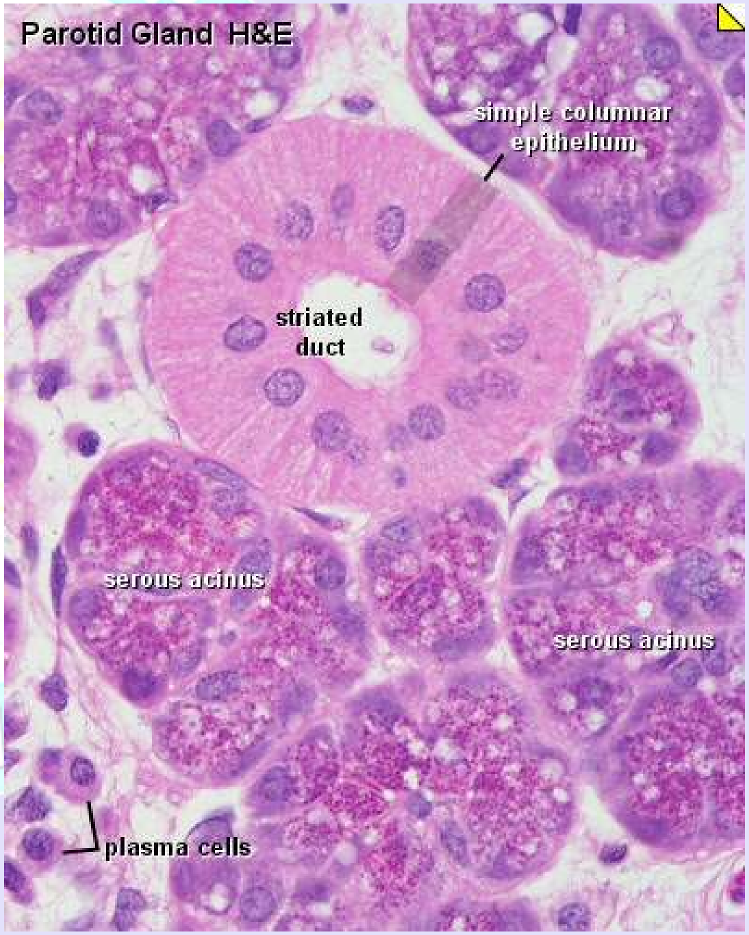


Embryology

- **6th-8th Weeks of Gestation**
- **Parotid**
- **First to develop**
- **Last to become encapsulated**

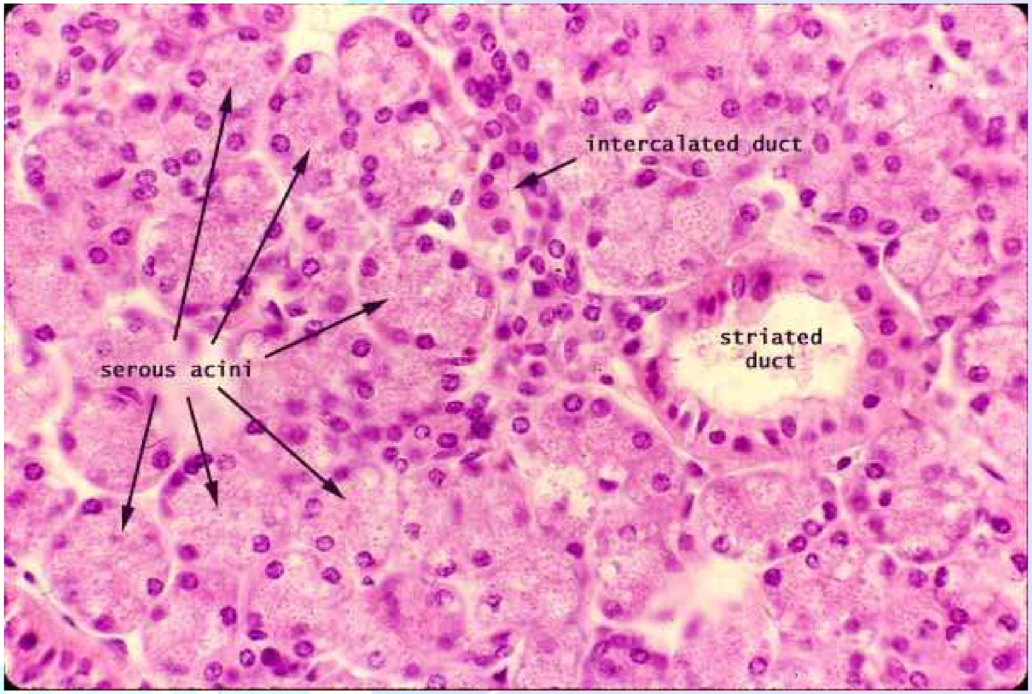
Parotid gland

- Largest.
- wrapped around the mandibular ramus, anterior and inferior to external ear
- Occupies the **parotid fascial space**
- Secretes saliva through Stensen's duct into the oral cavity, to facilitate mastication and swallowing.



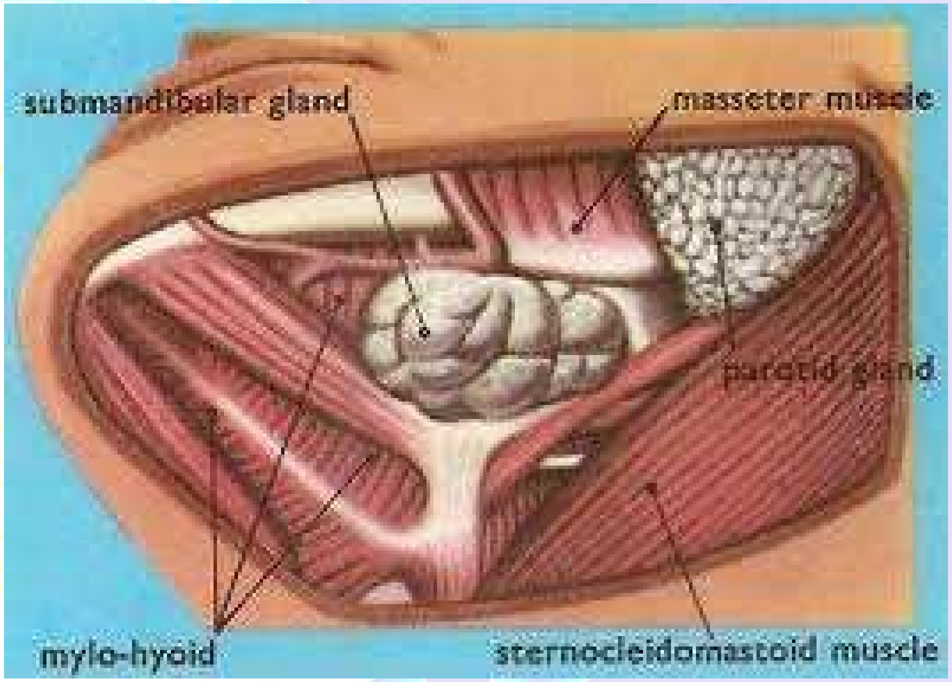
Parotid gland

- Serous acini only; contains numerous basophilic zymogen granules; nuclei are uniform, round and in basal half of cell
- Intercalated ducts are long but small in comparison to acini and striated ducts
- Striated ducts are larger than intercalated ducts, 3-6x size of acinus; striations are due to folds in basal plasma membranes
- Contains small lymph nodes near or within the gland, which arise from interstitial lymphocytes
- Resembles pancreatic tissue, but parotid gland has adipocytes and pancreatic tissue has islets and centroacinar cells



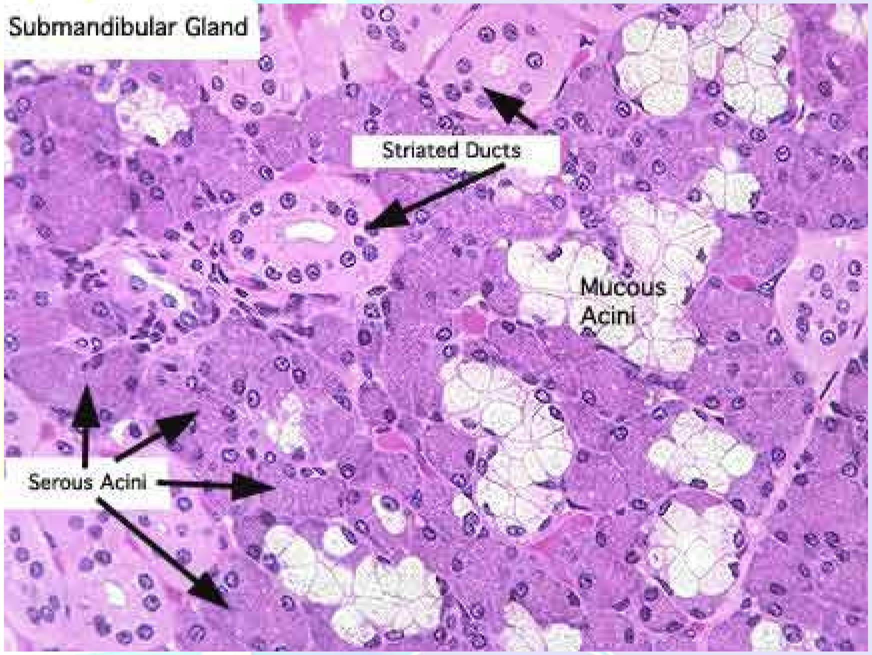
Submandibular gland

- Lying superior to the digastric muscles, each submandibular gland is divided into superficial and deep lobes.
- The superficial portion is larger.
- The deep portion is smaller.
- Secretions are delivered into the Wharton's ducts on the superficial portion after which they hook around the posterior edge of the mylohyoid muscle and proceed on the superior surface laterally.
- The ducts are then crossed by the lingual nerve, and ultimately drain into the sublingual caruncles on either side of the lingual frenulum along with the major sublingual duct (Bartholin).



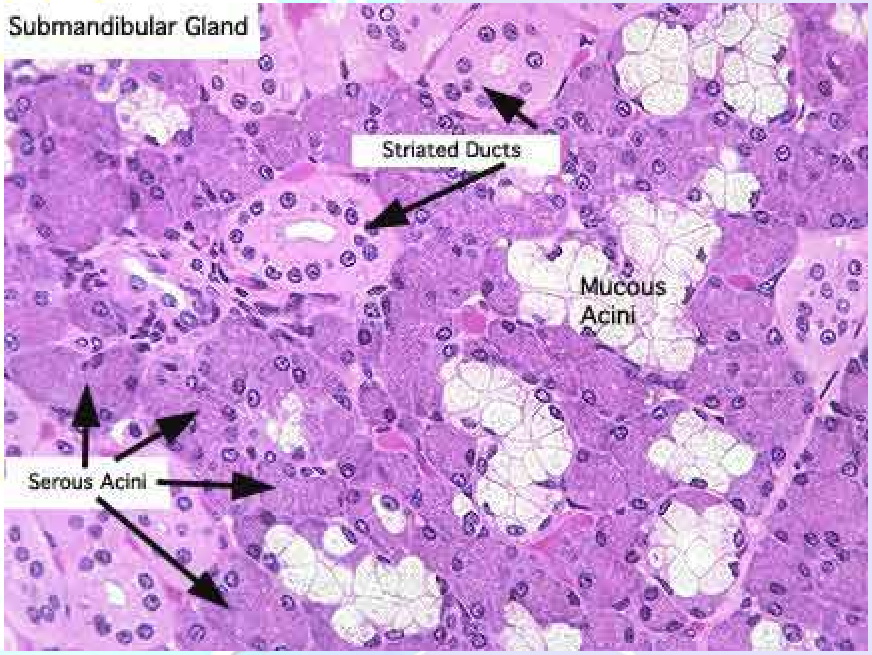
Submandibular gland

- Submandibular gland is a **compound tubuloalveolar gland**.
- It is surrounded by a capsule of moderately dense connective tissue, from which septa divide the gland into lobes and lobules.
- The secretory endpieces consist of a more or less spherical mass of cells called an **acinus**).



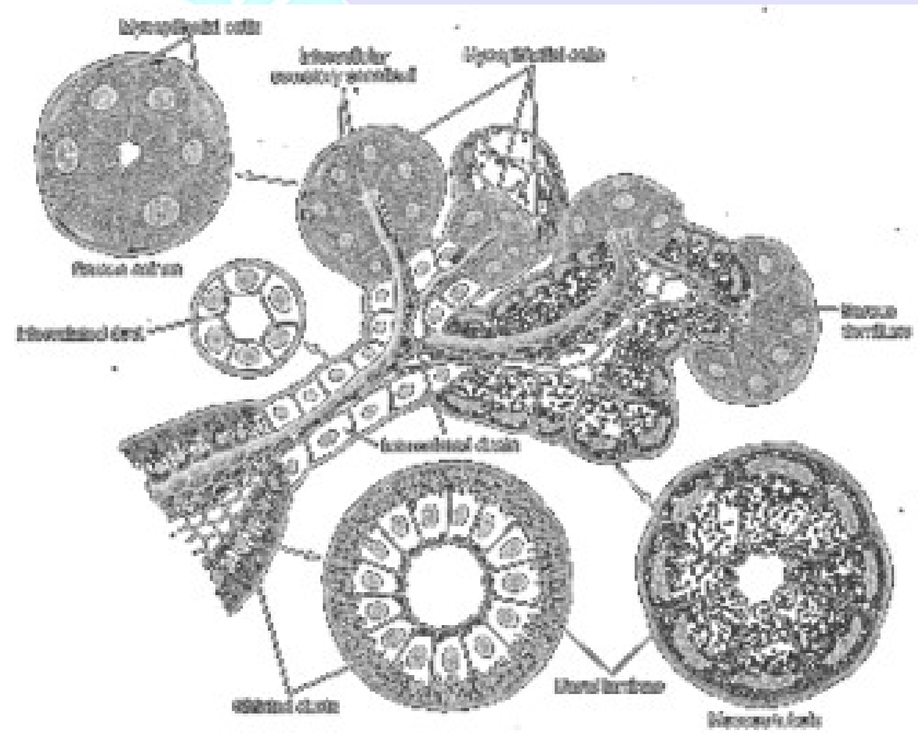
Submandibular glands

- Cells of the acini appear triangular in sections, with their apex directed toward the lumen, and their base resting on a basement membrane.
- They secrete their product in a merocrine fashion into the lumen.
- Contractile cells called **myoepithelial cells** or **basket cells** lie between the basement membrane and the plasma membrane of the secretory cells.
- They are also found in the proximal part of the duct system.
- Myoepithelial cells posess many actin-containing microfilaments, which squeeze on the secretory cells and move their products toward the excretory ducts.



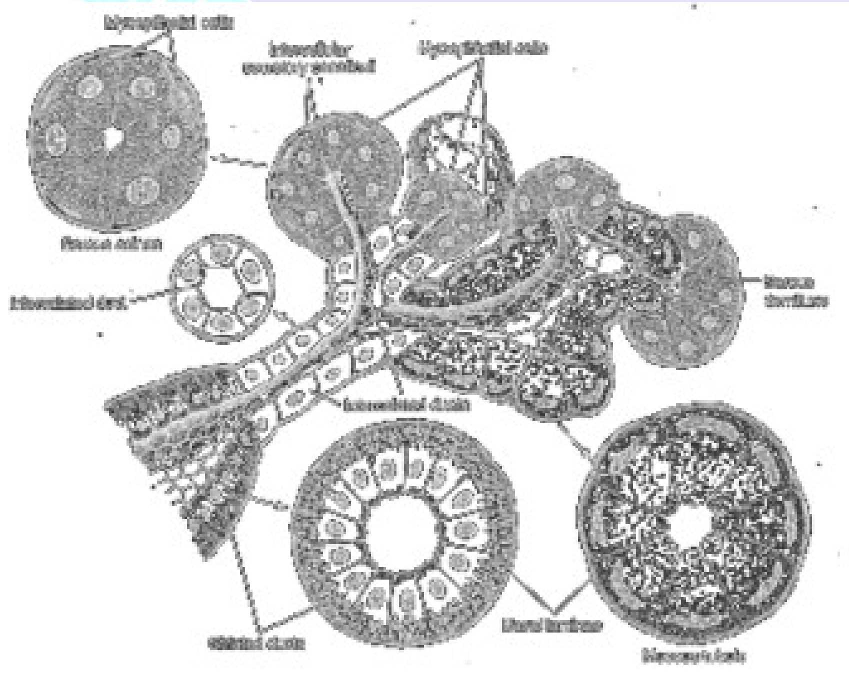
Submandibular gland

- Acini are
- **Serous**
  – secretion of serous cells is thin, watery and proteinaceous. Serous cells have a rounded nucleus and secretory granules in their cytoplasm. They are joined near their apical surfaces by junctional complexes
- **Mucous**.
  – Mucous cells secrete a viscous, glycoprotein-rich product, which is stored as mucinogen granules. The nuclei are typically flattened against the base of the cells (unless the cells have just discharged their contents, in which case they look more like serous cells).
  – Mucous cells typically look pale and empty in standard histological sections, because their granules are lost during preparation.
- The submandibular gland of humans is predominantly serous. Its mucous acini are quite frequently capped with a **serous demilune**, a crescent of serous cells around one or more of their surfaces.



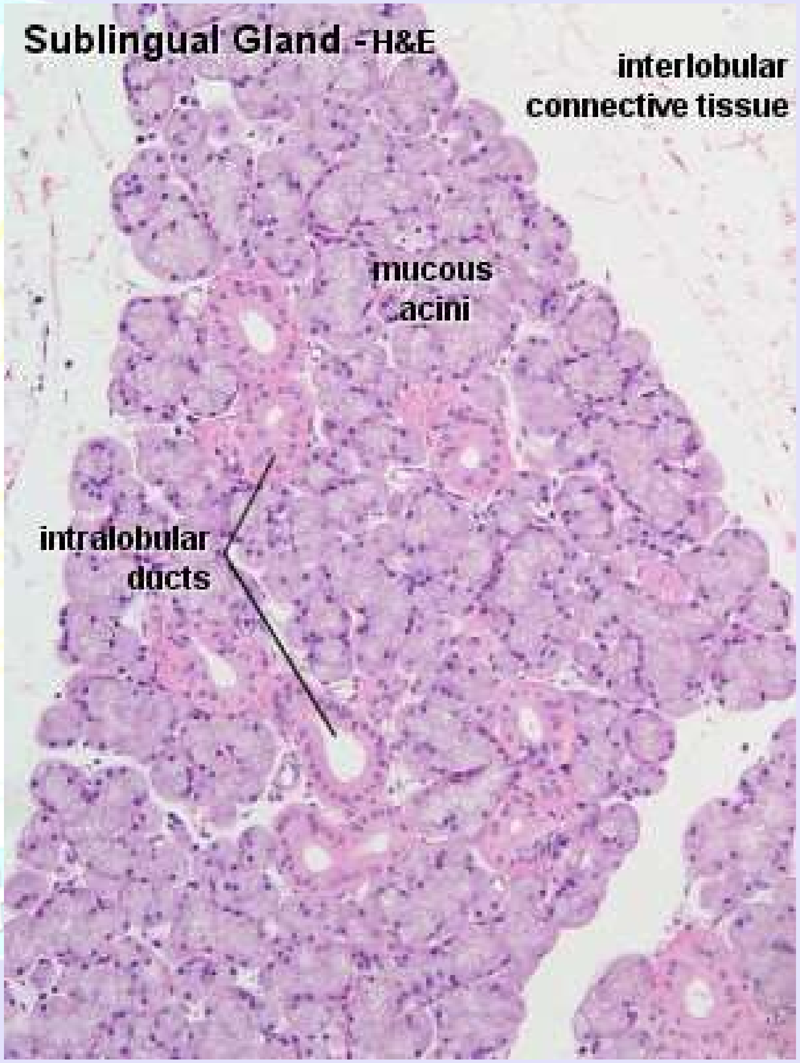
Submandibular gland

- 3 types of ducts in the submandibular gland:
- **Intercalated**
  - Intercalatd ducts are slender ducts continuous with the terminal acini, and lined with flat, spindle-shaped cells
- **Secretory** (**striated**)
  - Secretory ducts have eosinophilic cuboidal to columnar cells with basal striations. These result from infoldings of the basal membranes in which are found many mitochondria.
- Both intercalated and secretory ducts are found within the parenchyma of the gland and are therefore **intralobular ducts**.
- **Excretory**.
- The largest ducts are the excretory ducts. They are found in the connective tissue septa, and are therefore **interlobular ducts**. They ultimately connect with the oral cavity. Their epithelium is variable, it can be simple cuboidal, stratified cuboidal, stratified columnar or pseudostratified. Near the oral cavity, it becomes stratified squamous. Excretory ducts do not change the secretory product.



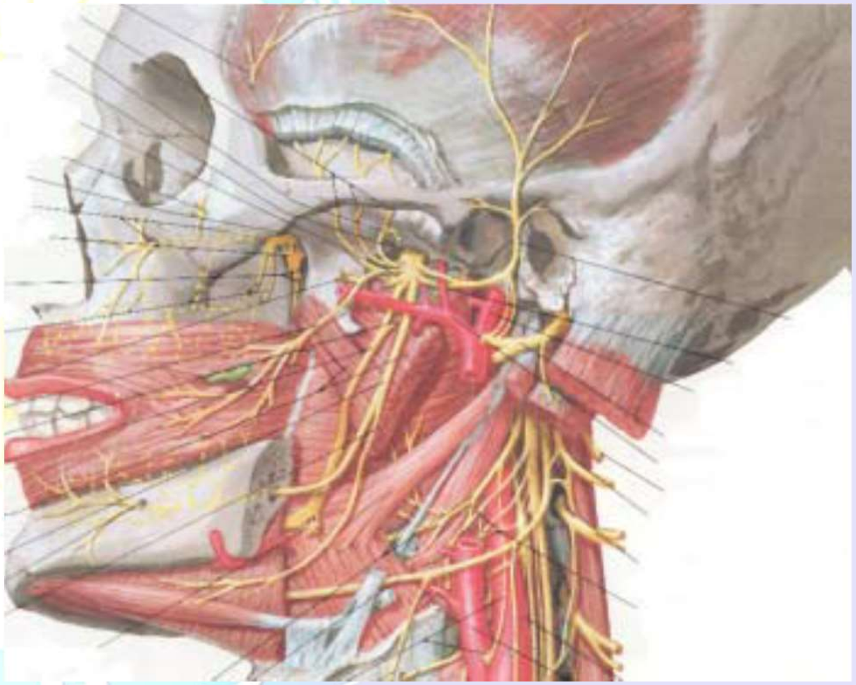
Sublingual gland

- They lie anterior to the submandibular gland under the tongue, beneath the mucous membrane of the floor of the mouth.
- They are drained by 8-20 excretory ducts called the ducts of Rivinus.
- The largest duct, the sublingual duct (of Bartholin) joins the submandibular duct to drain through the sublingual caruncle.
- The sublingual gland consists mostly of **Mucous** acini capped with serous demilunes and is therefore categorized as a mixed gland.
- Most of the remaining small sublingual ducts open separately into the mouth on an elevated crest of mucous membrane, the plica fimbriata, formed by the gland and located on either side of the frenulum linguae.
- The chorda tympani nerve (from the facial nerve via the submandibular ganglion) is secretomotor to the sublingual glands.



Autonomic innervation

- **Parasympathetic**
  - **Abundant,**
  - **watery saliva**
  - **Amylase down**
- **Sympathetic**
  - **Scant, viscous**
  - **saliva**
  - **Amylase up**



Thank you